Cedric Phillips

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Milan, IT | Canadian citizen

Education

- **Politecnico di Milano** *Fall 2026 – Spring 2028 (expected)*
  M.Sc. Mathematical Engineering — Statistical Learning Milan, Italy
  - Courses: Applied statistics, computational statistics, model identification & analysis, stochastic dynamical models, algorithms (in C++).

- **McGill University** *Fall 2022 – Spring 2026*
  B.Sc. Honours Applied Mathematics (First Class) Montreal, Canada
  - Minor concentrations: GIS & Remote Sensing, Hindi Language
  - GPA: 3.65/4.00 · Exchange at the École Polytechnique Fédérale de Lausanne (EPFL), Winter 2025
  - Courses: Probability theory, math. stats, stochastic processes, numerical analysis, algorithms, spatial stats.

Publications & Preprints

- **Two short proofs of incompatibility for correlation matrices** *2026*
  C. Phillips — arXiv:2609.14610 [math.ST]
  - Proved that some correlation matrices are not attainable with fixed margins: $d = 11$ for uniform margins (Spearman's rho) and $d = 6$ for arcsine margins, settling a conjecture of Devroye & Letac

Projects

- **Dependence modelling in aggregation trees (Honours Thesis)** *Winter–Summer 2026*
  Hierarchical copula models for correlated binary risks (e.g., credit defaults) Supervised by Prof. Nešlehová & Prof. Valiquette
  - Developed algorithms to learn the tree structure of hierarchical copula models for Bernoulli variables from correlation matrices, wrote Python code to test them for credit risk applications
  - Decomposed correlation as a Hadamard product over the tree, then built an equality-matching learning algorithm that doubled tree-recovery rates over a greedy baseline in simulation

- **Spatially Aware k-Nearest-Neighbours Regression** *January–April 2026*
  GEOG 506: Spatial Statistics Research Project Supervised by Prof. Sieber
  - Studied house-price prediction as nonparametric regression with spatially dependent errors; built a k-NN estimator with tunable spatial weights, selecting (k, λ) by validation MAE on ∼20k sales
  - Out-of-sample MAE fell 19% vs. the standard estimator (test R^2 0.73 → 0.78), also beating regression kriging; Moran's I confirmed residual spatial autocorrelation was absorbed

- **Stochastic optimization in an asymmetric board game** *Winter–Summer 2026*
  Maximal scoring strategies for the Lizard Cult in Root in The Delta Epsilon, McGill's undergraduate math journal
  - Modelled the game Root as a Markov chain on game states and one faction's action set; solved for expected-value-optimal play against varying adversaries, verified by Monte Carlo simulation in Python

Technical Skills & Proficiencies

- **Programming/data analysis:** Python (Pandas, NumPy, scikit-learn, matplotlib), SQL, R, Java
- **Statistical methods:** Copula models, Monte Carlo simulation, nonparametric regression, kriging
- **Languages:** English (native), French (native), Hindi-Urdu (advanced)

Experience

- **McGill University** *Winter 2025*
  Grader, MATH 458: Honours Differential Geometry Montreal, Canada
  - Sole grader for an advanced honours course; detailed written feedback on every assignment

- **Wild Blue Restaurant & Bar (North America's 50 Best)** *Summers & winters, 2021–2026*
  Expeditor Whistler, BC
  - Coordinated a continuous order flow under pressure; quality-checked every dish leaving the kitchen